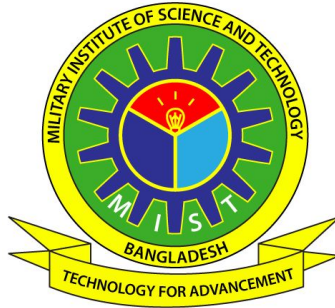


# MILITARY INSTITUTE OF SCIENCE AND TECHNOLOGY



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CSE-464: SOFTWARE DEVELOPMENT PROJECT-II

**PROJECT REPORT**

on

## **FoodRescue Sync**

Commercial Surplus Food Logistics

**Batch:** CSE-23    **Section:** A    **Group:** A6

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# 1 Problem Statement

Commercial food establishments generate waste massive amounts of food each day due to inefficiency and the absence of real-time logistics connecting them with local organizations or budget-conscious consumers. Surplus food that is still safe for consumption is discarded, contributing to lost revenue for businesses and land pollution. At the same time, a large portion of Bangladesh's urban population faces food insecurity, with no accessible way to benefit from the surplus available even close by.

The absence of a centralized, rapid-response platform means that potential food donors and recipients (who may be individuals or organizations) cannot coordinate efficiently. By the time food is identified as surplus, food begins to rot, making redistribution impractical. FoodRescue Sync [1] addresses this gap by providing a real-time logistics platform that connects commercial food donors with organizations and consumers to report surpluses and ensuring food reaches those in need before it is wasted.

## 2 Project Objectives

1. To develop a circular economy and supply-chain logistics platform that eliminates commercial food waste by connecting donors with consumers in real time.
2. To provide businesses with revenue recovery streams and easier donation facility.
3. To combat food insecurity of the poor people by enabling organizations and individuals to claim donations and purchase surplus food at a discounted price.

## 3 Finalized Features

1. **Inventory Management:** Donors can upload and maintain inventory via barcode scanning or manual entry, enabling fast and accurate stock tracking.
2. **Expiry Tracking:** The system monitors each item's expiry date and automatically applies a "surplus" tag when the threshold is approached, triggering redistribution workflows.
3. **Flash Sales:** Near-expiry items are listed on a discounted marketplace, giving donors an opportunity to recover sunk costs while minimizing waste.
4. **Donation Logging:** Completed donations are automatically recorded, and receipts are generated and stored for the donor's records.
5. **Surplus Radar:** A real-time map displays active surplus food listings in the user's vicinity, en-

abling consumers to locate available food quickly.

6. **Bulk Claiming:** Organizations can instantly claim large batches of donated food through a bulk-claim workflow designed for high-volume pickups.
7. **Discount Marketplace:** A consumer-focused marketplace lists discounted surplus from nearby restaurants and caterers for direct purchase.
8. **Custom Alerts:** Push notifications alert users about new listings matching their preferred food types or favorite establishments.

## 4 Entity-Relationship (ER) Diagram



Figure 1: Entity-relationship diagram for FoodRescue Sync

## 5 Tentative UI Pages

Table 1: Tentative UI pages and their descriptions

| Page                       | Description                                                                                                                   |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Welcome Screen             | Introduces the app and guides new users through initial role selection (donor, organization or consumer).                     |
| Login/Register             | Allows users to sign in to an existing account or create a new one with role-based registration.                              |
| Donor Dashboard            | Central hub for donors to view inventory summary, active entries and quick action shortcuts.                                  |
| Add/Scan Inventory         | Enables donors to add items via barcode scan or manual form with quantity and expiry date.                                    |
| Expiry Tracker             | Displays a sorted list of items approaching expiry with countdown timers.                                                     |
| Create Listing             | Allows donors to list a surplus item as a flash sale or donation with price and pickup window.                                |
| Listing Image Manager      | Provides donors with an interface to capture, upload, and preview food item photos during inventory listing.                  |
| Donation Log               | Shows a history of completed donations with status and downloadable tax-deductible receipts.                                  |
| Organization Dashboard     | Central hub for organizations showing available donation listings, active requests, and upcoming pickups.                     |
| Organization Profile Setup | Enables non-profit entities to register and manage their institutional details, verification status, and contact information. |
| Surplus Radar (Map)        | Real-time map of nearby active surplus food listings with type, distance, and category filters.                               |
| Bulk Request               | Allows organizations to select multiple donation listings and submit a consolidated request in one step.                      |
| Request Status Tracker     | Allows organizations and consumers to view the real-time approval status of their submitted food requests.                    |
| Pickup Coordination Screen | Displays pickup addresses, designated times, and logistical status updates for both donors and recipients.                    |
| Consumer Marketplace       | Displays discounted surplus meals from nearby vendors with search and category filters.                                       |

| Page                | Description                                                                                     |
|---------------------|-------------------------------------------------------------------------------------------------|
| Notification Centre | Shows personalised push notifications for new listings matching the user's saved preferences.   |
| Profile & Settings  | Allows users to update account details, food-type preferences, and alert notification settings. |

## 6 Tentative APIs

### 6.1 Authentication

Table 2: Authentication API endpoints

| Endpoint            | Description                                                                |
|---------------------|----------------------------------------------------------------------------|
| POST /auth/register | Register a new user with a chosen role (donor, organization, or consumer). |
| POST /auth/login    | Authenticate a user and return a session token.                            |
| POST /auth/logout   | Invalidate the current user's active session.                              |

### 6.2 Food Listings

Table 3: Food listings management API endpoints

| Endpoint                                                                          | Description                                                                 |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| POST /listings                                                                    | Create a new food listing with item details, expiry date, and listing type. |
| GET /listings?type={type}&category={category}&lat={lat}&lng={lng}&radius={radius} | Retrieve all active listings, filterable by type, category, or proximity.   |
| GET /listings/:id                                                                 | Fetch the full details of a single listing by ID.                           |
| PATCH /listings/:id                                                               | Update an existing listing's price, quantity, or status.                    |
| DELETE /listings/:id                                                              | Remove a listing once the food is no longer available.                      |
| GET /listings/nearby?lat={lat}&lng={lng}&radius={radius}                          | Return active listings within a given radius of the user's coordinates.     |

## 6.3 Requests

Table 4: Requests API endpoints

| Endpoint                                                            | Description                                                                 |
|---------------------------------------------------------------------|-----------------------------------------------------------------------------|
| POST /requests                                                      | Create a new request for a specific food listing.                           |
| GET /requests/:id                                                   | Retrieve the specific details and current status of a request.              |
| PATCH /requests/:id                                                 | Update a request's status (e.g., pending, accepted, or rejected).           |
| GET /requests/user/:userId                                          | Fetch all requests made by a specific user.                                 |
| GET /requests?userId={userId}&status={status}&listingId={listingId} | Fetch all requests, with optional filters by requester, status, or listing. |

## 6.4 Pickup Assignments

Table 5: Pickup assignments API endpoints

| Endpoint           | Description                                                                       |
|--------------------|-----------------------------------------------------------------------------------|
| POST /pickups      | Initialize a scheduled pickup tracking entry once a request is accepted.          |
| GET /pickups/:id   | Fetch logistics details, timing, and fulfillment tracking for a pickup.           |
| PATCH /pickups/:id | Update the operational status of a pickup (e.g., scheduled, en-route, completed). |

## 6.5 Food Images

Table 6: Food images API endpoints

| Endpoint                  | Description                                                  |
|---------------------------|--------------------------------------------------------------|
| POST /listings/:id/images | Upload and link a photo asset to a specific food listing ID. |
| DELETE /images/:id        | Remove an uploaded food image asset from the system.         |



## 6.6 Organizations

Table 7: Organizations API endpoints

| Endpoint               | Description                                                             |
|------------------------|-------------------------------------------------------------------------|
| POST /organizations    | Create an institutional organization profile associated with a user ID. |
| GET /organizations/:id | Retrieve profile details for a verified organization.                   |

## 6.7 Donation Logs

Table 8: Donation logs API endpoints

| Endpoint                                                       | Description                                                            |
|----------------------------------------------------------------|------------------------------------------------------------------------|
| POST /donations                                                | Log a completed donation and trigger automated tax receipt generation. |
| GET/donations/{userId}?startDate={startDate}&endDate={endDate} | Retrieve all donation records and receipts for a given donor.          |

## 6.8 Notifications

Table 9: Notifications API endpoints

| Endpoint                                                 | Description                                                                            |
|----------------------------------------------------------|----------------------------------------------------------------------------------------|
| GET/notifications/{userId}?limit={limit}&offset={offset} | Fetch all notifications for a specific user, ordered from most recent to least recent. |
| PATCH /notifications/:id                                 | Mark a specific notification as read.                                                  |

## 7 App Reviews

Each group member reviewed one existing food rescue or surplus food platform to identify best practices and some key gaps that FoodRescue Sync hopes to address.

## 7.1 Too Good To Go (Reviewed by: Lt Mashfiq Ahmed Chowdhury)

Too Good To Go [2] is a commercial surplus food marketplace operating in over 17 countries that connects consumers with restaurants and stores selling near-expiry food in discounted “Magic Bags.”

Table 10: Too Good To Go: features and limitations

| Sl. No. | Features                                                                                      | Limitations                                                                    |
|---------|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| 1       | Magic Bag system: ”bags” or packages of near-expiry food sold at a significant fixed discount | No organization or charity donation module. unsuitable for bulk redistribution |
| 2       | Location-based map showing partner restaurants and stores                                     | Individual item selection not possible, only ”bags” available                  |
| 3       | Secure in-app payment processing for consumers                                                | Inventory management and expiry tracking not available to donors               |
| 4       | Personal impact dashboard showing meals saved and carbon dioxide reduced                      | Not operational in Bangladesh                                                  |
| 5       | User ratings and reviews                                                                      | No specialized push notification system                                        |

## 7.2 OLIO (Reviewed by: Lt Mehedi Hasan Redoy)

OLIO [3] is a UK-based community sharing platform for free food and household item exchange. Its Food Waste Heroes program enlists volunteers to collect unsold surplus food from supermarkets and redistribute it locally.

Table 11: OLIO: features and limitations

| Sl. No. | Features                                                                          | Limitations                              |
|---------|-----------------------------------------------------------------------------------|------------------------------------------|
| 1       | Free food and item sharing between community members using posts with photo       | No expiry tracking for businesses        |
| 2       | Location based matching of listings                                               | No automated donation receipt generation |
| 3       | Food Waste Heroes: volunteer driven supermarket extra food redistribution program | All transactions are free, no profiting  |
| 4       | In-app direct messaging between listers and recipients                            | No bulk claiming feature                 |

| Sl. No. | Features                                                                                 | Limitations                      |
|---------|------------------------------------------------------------------------------------------|----------------------------------|
| 5       | Reduced Food section showing realtime discounted grocery items from nearby supermarkets. | Platform dependent on volunteers |

### 7.3 Food Rescue US (Reviewed by: Effat Jahan Erin)

Food Rescue US [4] is a US-based non-profit platform that matches food donors (restaurants, caterers, grocers) with registered non-profit recipient agencies. In doing this, the platform uses trained volunteers known as “Food Runners” for pickups.

Table 12: Food Rescue US: features and limitations

| Sl. No. | Features                                                                         | Limitations                                                                  |
|---------|----------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| 1       | Direct matching of verified food donors with registered non-profit organizations | Completely volunteer dependent                                               |
| 2       | Real-time donation status tracking for donors, agencies, and volunteers          | No inventory management, expiry tracking, or automated surplus-tagging tools |
| 3       | Total impact reporting in dashboard                                              | No flash sale feature                                                        |
| 4       | Volunteer scheduling and management system                                       | Only available in the United States                                          |

### 7.4 Flashfood (Reviewed by: Wasif Darain)

Flashfood [5] is a North American grocery surplus marketplace that partners with major supermarket chains to list near-expiry grocery items at discounts of up to 50%, available for in-store pickup.

Table 13: Flashfood: features and limitations

| Sl. No. | Features                                               | Limitations                                                                         |
|---------|--------------------------------------------------------|-------------------------------------------------------------------------------------|
| 1       | Store pickup for groceries near expiry at 50% discount | Covers grocery stores only. Does not support restaurants, caterers and event venues |
| 2       | Wide partnership network in the US                     | Service only available in the United States                                         |
| 3       | QR code-based pickup verification at the store counter | No real-time expiry tracking                                                        |

| Sl. No. | Features                                                        | Limitations                                                             |
|---------|-----------------------------------------------------------------|-------------------------------------------------------------------------|
| 4       | Consumer savings tracker that shows savings                     | The platform is purely commercial                                       |
| 5       | In-app notification system for new listings in the user's area. | No alerting for listings. Users must browse to discover available items |

In all, these apps show the viability of food waste reduction mobile apps and provide good user interfaces (UI). Across all platforms, critical features are missing, such as expiry tracking, inventory management tools, bulk claiming for organizations and customized alert systems. In addition, some of these apps are purely volunteer driven and some are purely commercial and none are available in Bangladesh, so their action in practice could not be observed. FoodRescue Sync hopes to fill most of these gaps, especially in a Bangladeshi context.

## 8 Contribution Table

Table 14: Group member contributions

| ID No.    | Member Name                | Contribution                                                |
|-----------|----------------------------|-------------------------------------------------------------|
| 202114191 | Lt Mashfiq Ahmed Chowdhury | Problem Statement, App Review (Too Good To Go)              |
| 202114196 | Lt Mehedi Hasan Redoy      | Project Objectives, Finalized Features, App Review (OLIO)   |
| 202314020 | Effat Jahan Erin           | ER Diagram, Tentative UI Pages, App Review (Food Rescue US) |
| 202314053 | Wasif Darain               | Report Compilation, Tentative APIs, App Review (Flashfood)  |

## 9 References

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